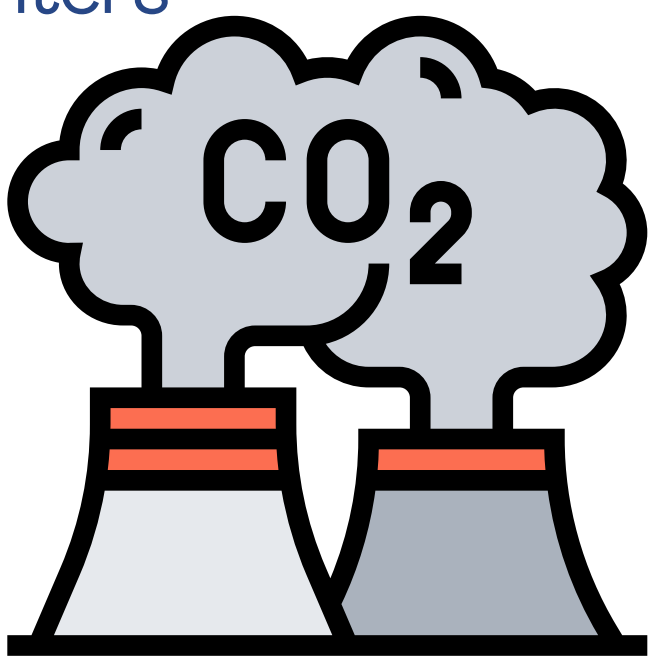
Environmental Concerns

Data Centers



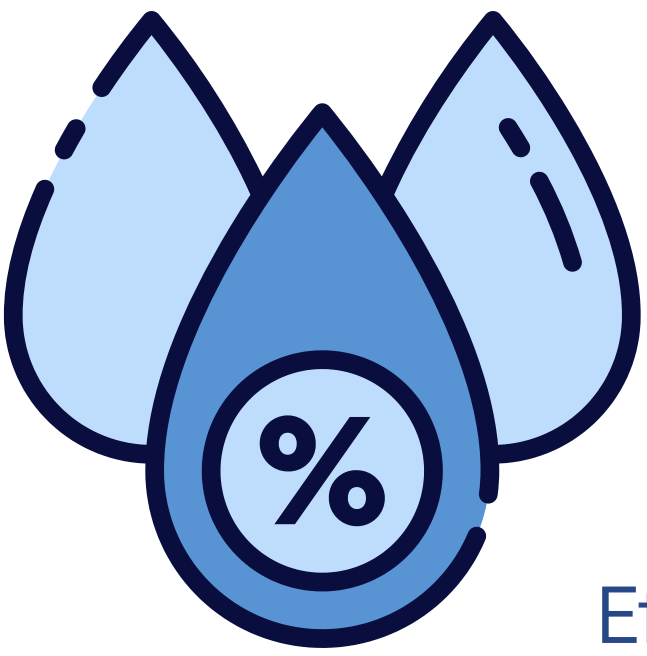
Data centers have taken up more land in total than the Jakarta Metropolitan area. These data centers do not simply take up more land, but more energy as well. As these data centers need more energy, the use of fossil fuels has returned, increasing carbon footprints. The United Nations has stated that data centers may consume 945 terawatt-hours of electricity every year starting from 2030. This is triple the summed annual electricity consumption of the countries Pakistan, Bangladesh and Nigeria.

Developers are also increasingly targeting open green spaces to build data centers. Such locations include rural areas, farm lands, and green spaces such as forests. The exploitation of these lands cause the fragmentation of wild life habitats and various other issues.

Water Supply

Other than land and energy, one of the largest concerns with data centers is the water consumption.
Water is crucial in data centers, since there is cooling required for all the computing that the AI models are doing. An international water scarcity issue had already existed before Artificial Intelligence started demanding so much water, and there are already places in the world that have a limited supply for drinkable water.

A single query with AI uses a small amount of water, but with billions of users using AI chatbots everyday, the number builds up to massive amounts. In addition, facilities often draw from local water supplies, even during drought season.



AI and water consumption has become an increasingly discussed issue, with different numbers coming from various differen sources regarding the actual use of water.

Efforts

Efforts regarding conservation and sustainability have taken various approaches. Some claim that AI will help conservation efforts in return by optimizing processes in industries. On the other hand, some claim that AI should take a back seat priority in order to further conservation efforts and renewable energy.